SBE37-SMP-ODO MicroCAT

Instrument Configuration

Instrument Serial Number:	37-16256
Instrument Firmware Version:	2.13.0
Zero Conductivity Frequency:	2806.31
Communications Format:	RS232
Communications Settings:	9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

Data Format	*Measurement*	*Sensor Type*	*Serial Number*	*Rating*
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10723494	2000m(2000 dBar)
RS232	Oxygen	SBE 63	63-1802	7000m

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.

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SENSOR SERIAL NUMBER: 16256
CALIBRATION DATE: 06-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

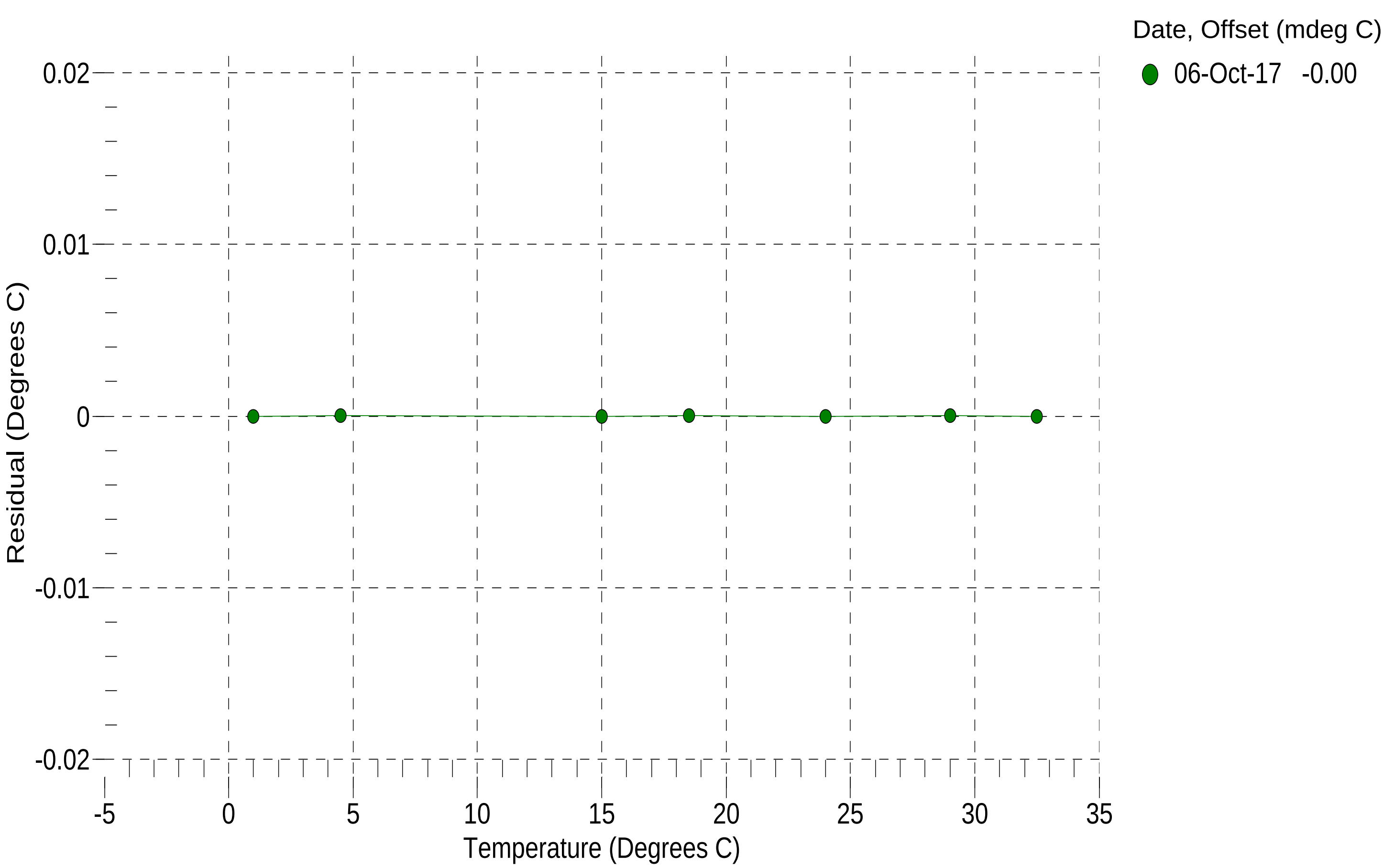
COEFFICIENTS:
a0 = -1.581247e-004
a1 = 3.126029e-004
a2 = -4.630448e-006
a3 = 2.055773e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	565206.3	1.0000	-0.0000
4.5000	484287.9	4.5000	0.0000
15.0000	310644.0	15.0000	-0.0000
18.5000	269571.1	18.5000	0.0000
24.0000	217004.9	24.0000	-0.0000
29.0000	179251.5	29.0000	0.0000
32.5000	157324.7	32.5000	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature



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SENSOR SERIAL NUMBER: 16256
CALIBRATION DATE: 06-Oct-17

SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -9.829126e-001
h =  1.248619e-001
i = -7.759067e-005
j =  2.090562e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC = -1.1877e-007
```

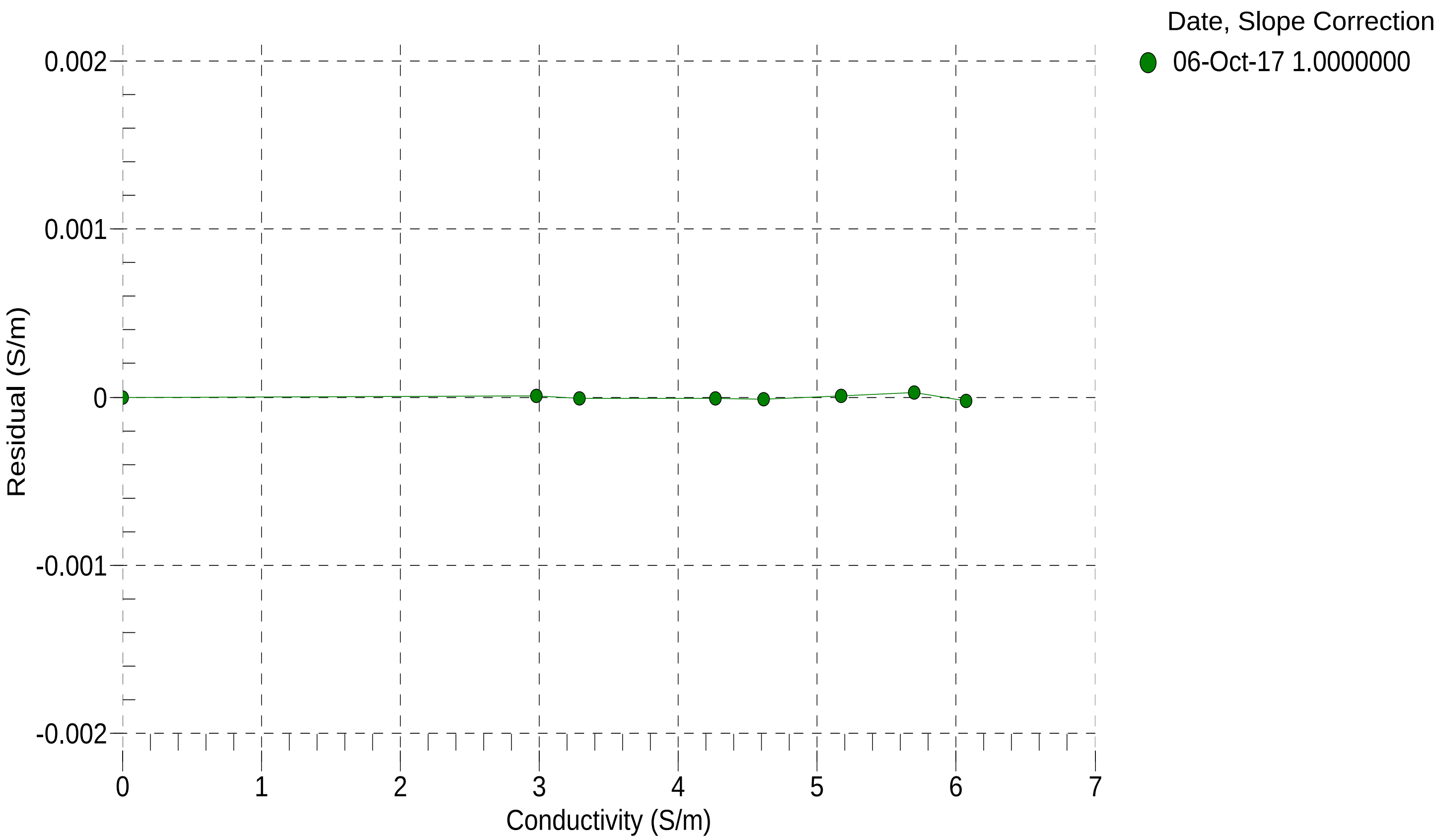
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2806.31	0.00000	0.00000
1.0000	34.8645	2.97969	5628.40	2.97970	0.00001
4.5000	34.8444	3.28711	5841.85	3.28711	-0.00001
15.0000	34.8011	4.26995	6476.52	4.26994	-0.00001
18.5000	34.7920	4.61549	6685.08	4.61548	-0.00001
24.0000	34.7819	5.17407	7008.83	5.17408	0.00001
29.0000	34.7764	5.69652	7298.34	5.69655	0.00003
32.5000	34.7735	6.06937	7497.91	6.06935	-0.00002

f = Instrument Output(Hz) * sqrt(1.0 + WBOTC * t) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 16256
CALIBRATION DATE: 03-Oct-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10723494

COEFFICIENTS:

PA0 = -2.371812e-001	PTCA0 = 5.242121e+005
PA1 = 8.954203e-003	PTCA1 = 2.767879e-001
PA2 = -1.421826e-010	PTCA2 = -6.507895e-002
PTEMPA0 = -5.883026e+001	PTCB0 = 2.498938e+001
PTEMPA1 = 5.438361e-002	PTCB1 = -5.250000e-004
PTEMPA2 = -7.315508e-007	PTCB2 = 0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.75	525859.3	1530.6	14.76	0.00
591.85	590336.4	1532.2	591.76	-0.00
1169.09	654974.4	1533.1	1169.01	-0.00
1746.23	719740.1	1533.7	1746.21	-0.00
2323.28	784626.2	1534.5	2323.28	0.00
2900.23	849621.5	1535.3	2900.13	-0.00
2323.26	784642.9	1534.8	2323.43	0.01
1746.14	719737.7	1534.0	1746.19	0.00
1169.10	654988.0	1533.6	1169.13	0.00
591.89	590344.0	1533.2	591.83	-0.00
14.75	525866.4	1533.2	14.83	0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1719	525901.35
29.00	1652	525917.94
24.00	1556	525934.72
18.50	1450	525945.68
15.00	1383	525949.82
4.50	1183	525963.91
1.00	1117	525963.25

TEMPERATURE (°C)	SPAN (mV)
-5.00	24.99
35.00	24.97

y = thermistor output (counts)

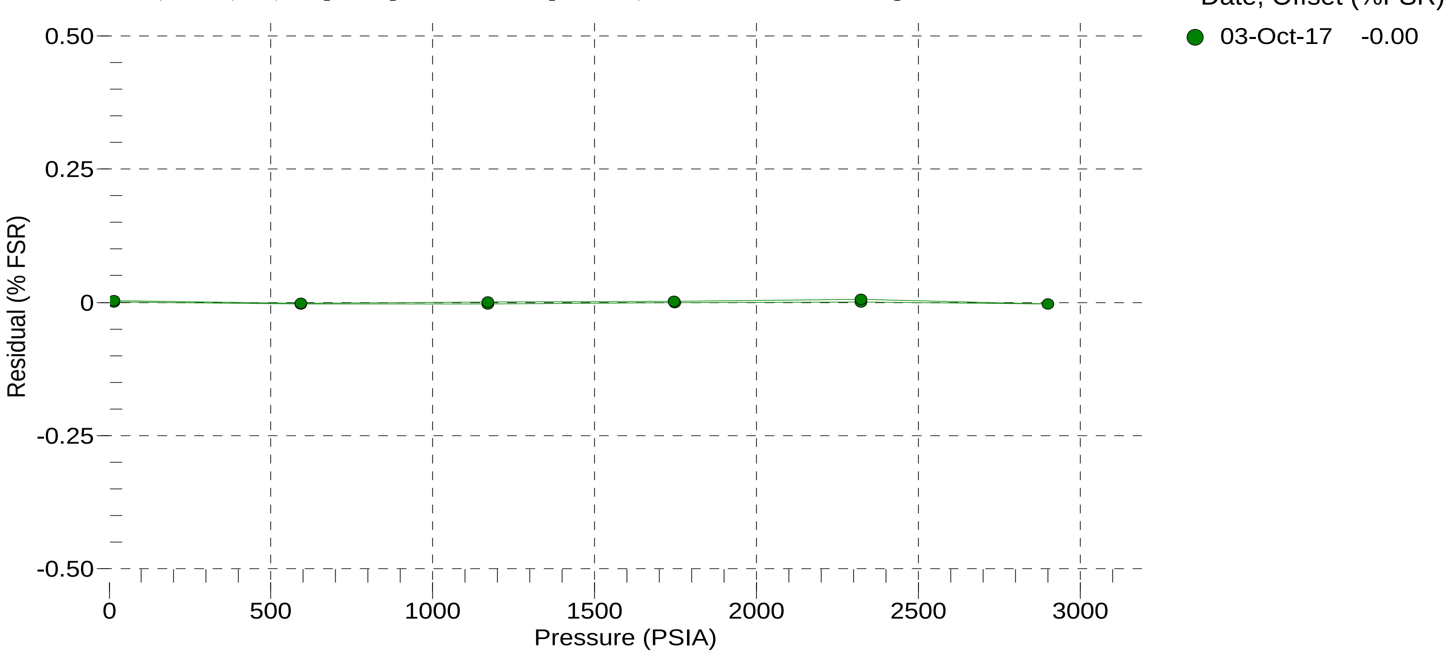
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2017-09-29

Description: SBE-37 Microcat

Sensor Information:

Model Number: SBE-37

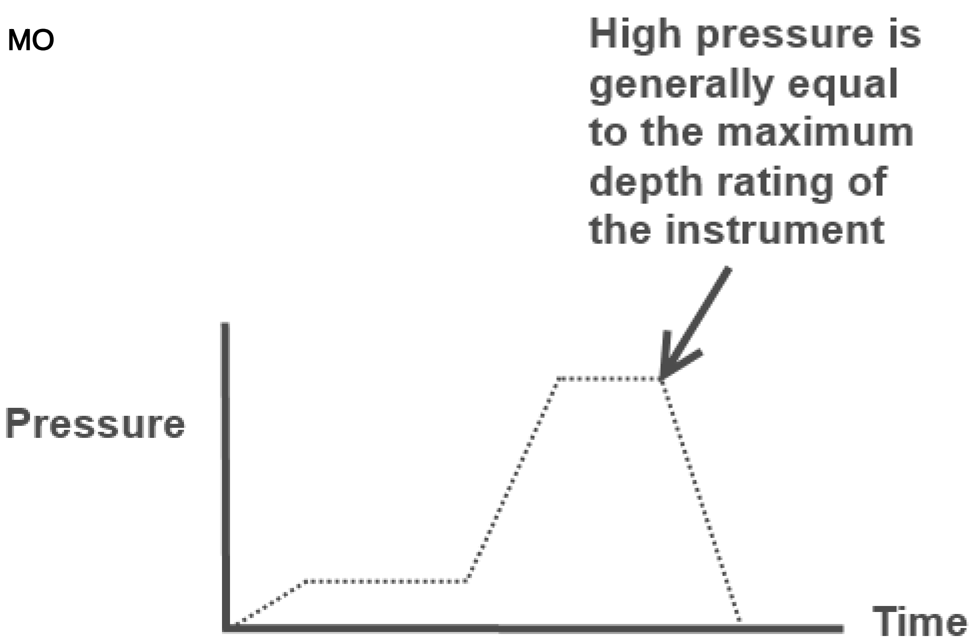
Serial Number: 16256

Pressure Test Protocol:

Low Pressure Test: 40	PSI	Held For:	15	Minutes
High Pressure Test: 2900	PSI	Held For:	15	Minutes

Passed Test: True

Tested By: MO



Typical Test Profile

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SENSOR SERIAL NUMBER: 1802
CALIBRATION DATE: 14-Sep-17

SBE 63 OXYGEN TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

TA0 = 6.810451e-004 TA2 = 5.920027e-008
TA1 = 2.574392e-004 TA3 = 1.213807e-007

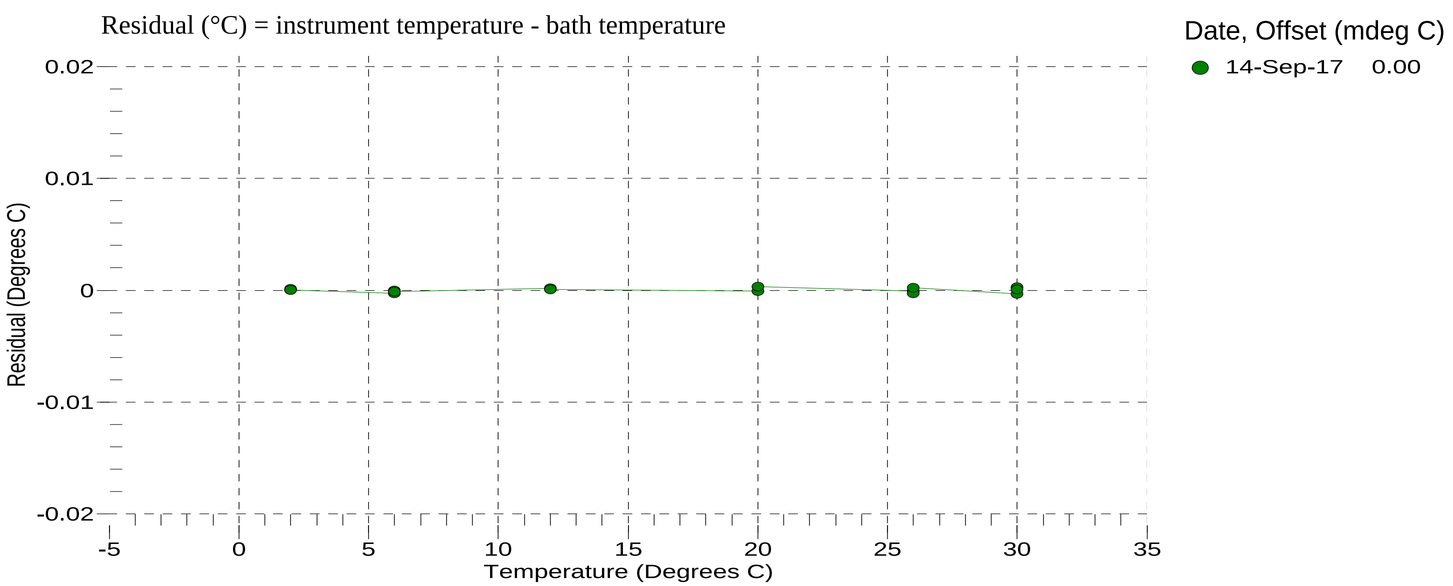
BATH TEMP (° C)	INSTRUMENT OUTPUT(V)	INST TEMP (° C)	RESIDUAL (° C)
1.9998	1.11765	1.9998	0.00003
2.0000	1.11764	2.0001	0.00014
2.0001	1.11764	2.0001	0.00004
2.0001	1.11764	2.0001	0.00004
5.9999	0.99349	5.9996	-0.00028
6.0000	0.99348	6.0000	-0.00004
6.0001	0.99348	6.0000	-0.00014
6.0001	0.99348	6.0000	-0.00014
12.0000	0.82771	12.0002	0.00017
12.0000	0.82771	12.0002	0.00017
12.0001	0.82771	12.0002	0.00007
12.0001	0.82771	12.0002	0.00007
20.0000	0.64420	19.9999	-0.00008
20.0000	0.64420	19.9999	-0.00008
20.0000	0.64420	19.9999	-0.00008
20.0001	0.64419	20.0004	0.00031
25.9999	0.53233	25.9998	-0.00009
26.0001	0.53233	25.9998	-0.00029
26.0001	0.53233	25.9998	-0.00029
26.0002	0.53232	26.0004	0.00020
30.0000	0.46860	29.9997	-0.00031
30.0001	0.46859	30.0004	0.00026
30.0001	0.46859	30.0004	0.00026
30.0003	0.46859	30.0004	0.00006

V = Instrument Output (Volts)

L = *ln* (100000 * V / (3.3 - V))

Temperature ITS-90 (°C) = 1 / (TA0 + (TA1 * L) + (TA2 * L^2) + (TA3 * L^3)) - 273.15

Residual (°C) = instrument temperature - bath temperature



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SENSOR SERIAL NUMBER: 1802

CALIBRATION DATE: 14-Sep-17

SBE 63 OXYGEN CALIBRATION DATA

COEFFICIENTS:

A0 = 1.0513e+000	B0 = -2.5690e-001	C0 = 1.0152e-001	E = 1.1000e-002
A1 = -1.5000e-003	B1 = 1.6680e+000	C1 = 4.3181e-003	
A2 = 4.3109e-001		C2 = 5.9460e-005	

BATH OXYGEN (ml/l)	BATH TEMPERATURE (° C)	BATH SALINITY (PSU)	INSTRUMENT OUTPUT (µsec)	INSTRUMENT OXYGEN (ml/l)	RESIDUAL (ml/l)
0.697	30.00	0.00	31.33	0.693	-0.004
0.723	26.00	0.00	31.98	0.720	-0.003
0.776	20.00	0.00	32.92	0.777	0.001
0.872	12.00	0.00	34.21	0.873	0.001
0.981	6.00	0.00	35.09	0.986	0.005
1.077	2.00	0.00	35.66	1.086	0.010
2.174	30.00	0.00	22.92	2.171	-0.003
2.306	26.00	0.00	23.45	2.305	-0.002
2.457	20.00	0.00	24.57	2.457	0.001
2.977	12.00	0.00	25.45	2.970	-0.007
3.386	6.00	0.00	26.40	3.378	-0.008
3.640	30.00	0.00	18.92	3.639	-0.002
3.769	2.00	0.00	26.97	3.760	-0.009
3.878	26.00	0.00	19.38	3.879	0.001
4.294	20.00	0.00	20.11	4.303	0.009
5.062	12.00	0.00	21.12	5.059	-0.003
5.179	30.00	0.00	16.42	5.172	-0.006
5.566	26.00	0.00	16.76	5.565	-0.001
5.804	6.00	0.00	21.96	5.801	-0.003
6.176	20.00	0.00	17.39	6.192	0.015
6.417	2.00	0.00	22.54	6.419	0.002
7.250	12.00	0.00	18.34	7.244	-0.006
8.295	6.00	0.00	19.11	8.296	0.001
8.824	2.00	0.00	19.96	8.829	0.005

T = temperature (°C) , P = pressure (dbar), U = Instrument output (µsec)

S_{corr} (salinity correction function) = 1.0 for calibration in DI water

See the user manual for more information on S_{corr} calculation

V = U / 39.457071

$$\text{Oxygen (ml/l)} = \{((A0 + A1*T + A2 * V^2)/(B0 + B1 * V) - 1.0)/(C0 + C1 * T + C2 * T^2)\} * S_{corr} * \exp(E * P / (T + 273.15))$$

Residual (ml/l) = instrument oxygen - bath oxygen